

Key Words: Arteries, capillaries, veins, valves, heart, circulation

Starter: A person wanted to measure how much oxygen was in the blood during different activities using an **oximeter**.

1. Suggest 3 activities they could get their participants to do

2. What is the **independent** and **dependent variable** of this experiment?

3. Describe what you would expect to happen as the intensity of activity increased. Explain why?



Review

1. Suggest 3 activities they could get their participants to do

Walking, jogging, running



2. What is the **independent and dependent variable** of this experiment?

I: Type of exercise

D: Oxygen percentage

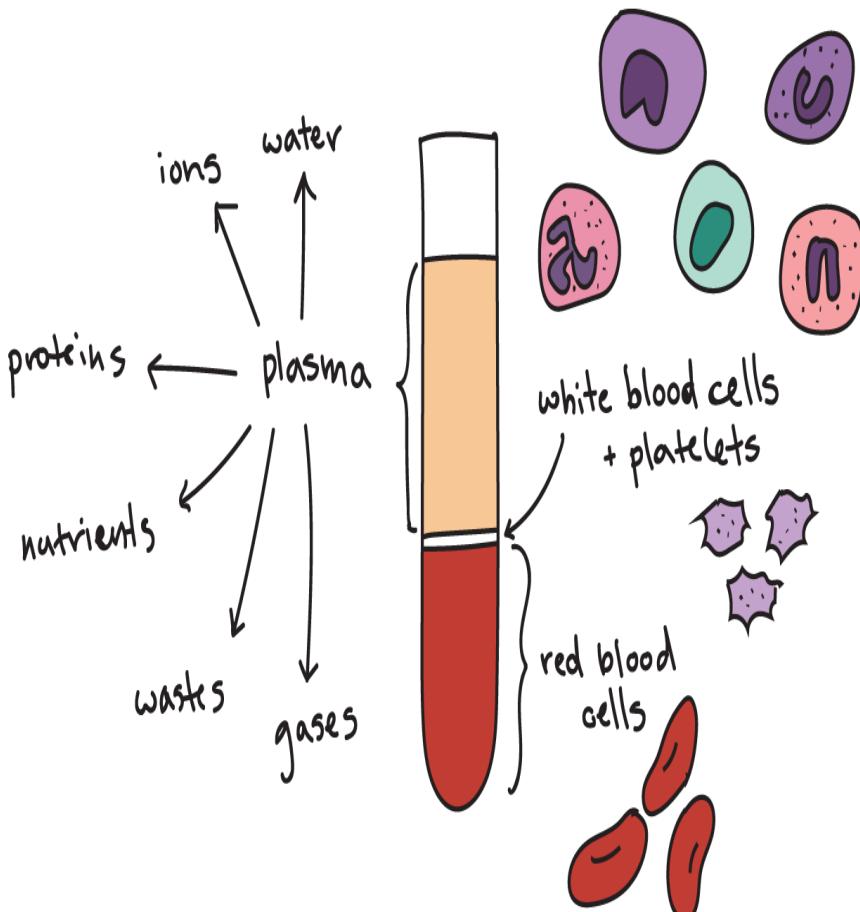
3. Describe what you would expect to happen as the intensity of activity increased. Explain why?

There isn't much change because as oxygen demand increases your body adapts by increasing breathing rate to keep your blood oxygenated for good supply to the working muscles.

Key Words: Arteries, capillaries, veins, valves, heart, circulation

DO NOW: Answer these three review questions below.

EXT: Name the four reagents used for food tests and their positive result.



1. Name two types of white blood cell. [1 mark]

2. Name a gas, waste product, nutrient, protein and ion that plasma may carry. [5]

3. Explain the function and adaptations of red blood cells [4]

Review

1. Name two types of white blood cell. [1 mark]

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Lymphocytes
Phagocytes

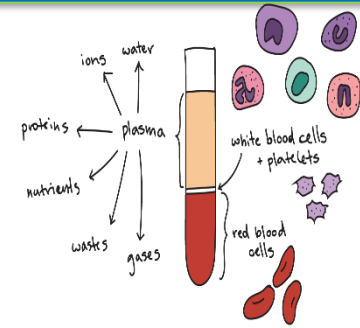
Gas = oxygen

Waste = urea/CO₂

Nutrient = glucose

Protein = hormone

Ion = Na⁺ K⁺ Mg⁺



Red blood cells carry oxygen. [1] They have a Molecule called haemoglobin that helps them to carry it. [1] They have no nucleus and a bio-concave shape [1]. This helps them pack in as much haemoglobin as they can. [1]

EXT: Name the four reagents used for food tests. And their positive result.

Iodine → starch Benedicts → sugar Biuret → Protein Ethanol + water → lipids

LOb: Understand structures associated with the circulation of blood through the body.

Keywords: Arteries, capillaries, veins, valves, heart, circulation

Success Criteria:

- Recall the structures of the blood vessels
- Explain adaptations that help blood vessels carry blood
- Assess the importance of the double circulatory system

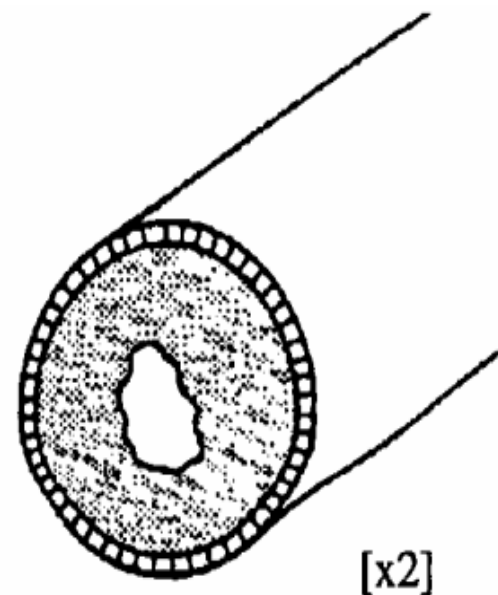
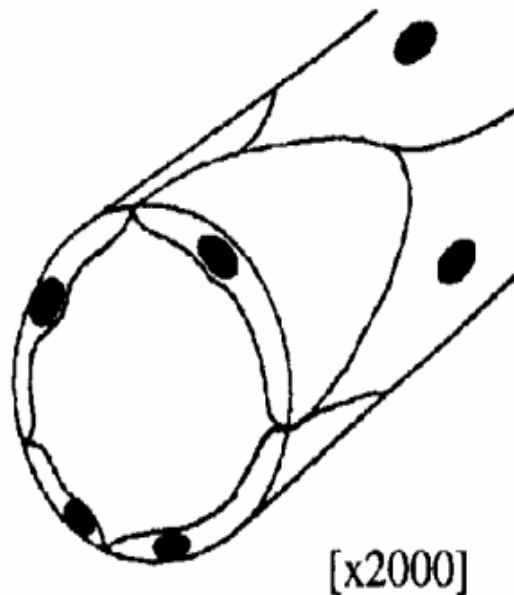
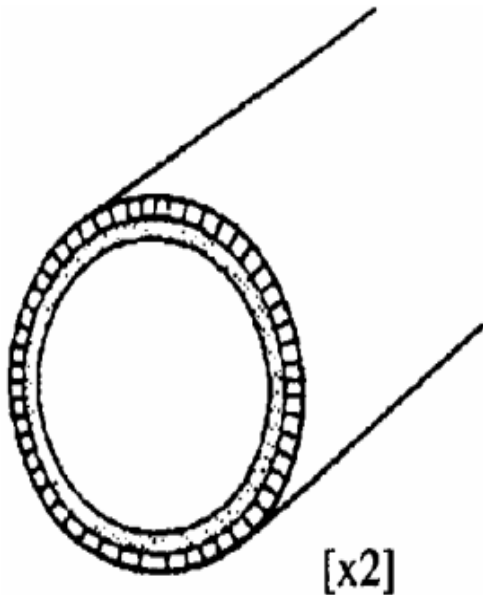


Did you know... the average human has about 5.6 litres of blood. The blood circulates round the body three times every minute and in one day a single blood cell travels 12,000 miles. 4x the distance of the USA coast to coast.

Blood Vessels

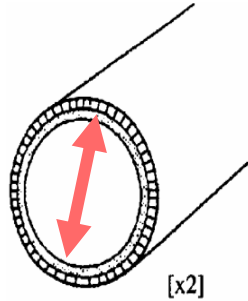
Blood vessels are the channels for blood to flow. Blood only travels in one direction and there are three different vessel types you need to know:

Arteries | Veins | Capillaries



Below will appear some information. If you know the vessel name 100% confidently write the name into your book.
Then turn your book over.

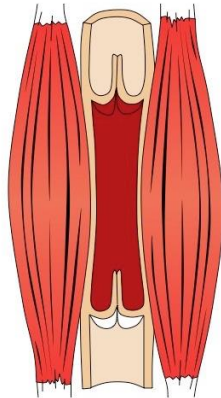
1.



Wide lumen

2.

Has valves



3.

Deoxygenated blood

4.

Low pressure blood

5.

Thinner walls
Thin muscle layer

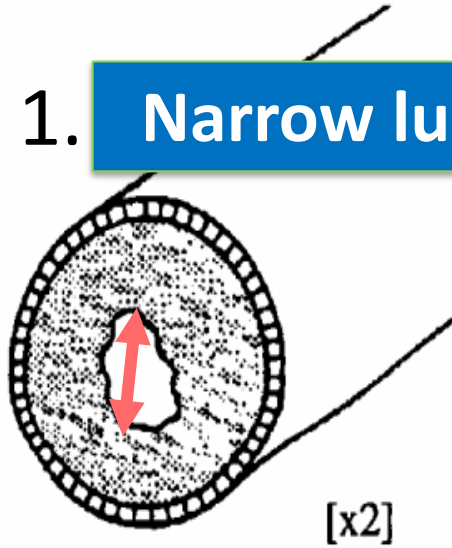
6.

Close to the surface
of the skin

7.

Takes blood too the
heart

Arteries have opposite structural properties to veins.



1. **Narrow lumen**

2. **No valves**

3. **Oxygenated blood**

4. **High pressure blood**

5. **Thick walls**
Thick muscle layer

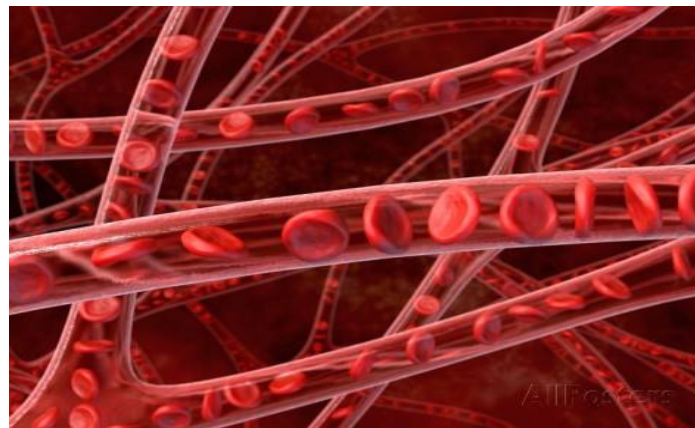
6. **Deep under the skin**

7. **Takes blood AWAY**
from the heart

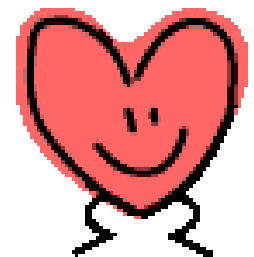
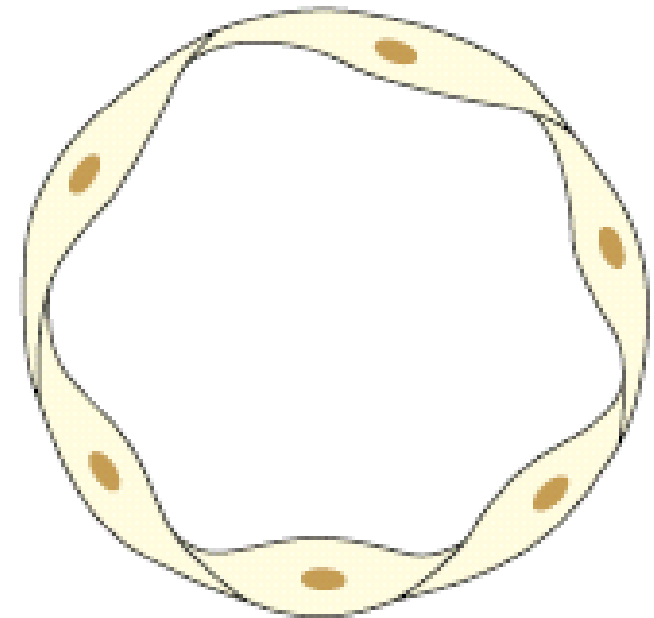
Capillaries carry blood at low pressure

Both oxygenated and deoxygenated

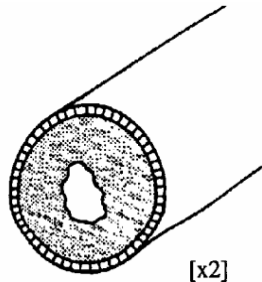
- No muscle
- No elastic
- No valves
- Very narrow
- Thin layer of one cells thick



Capillaries

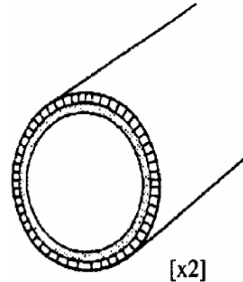


Artery



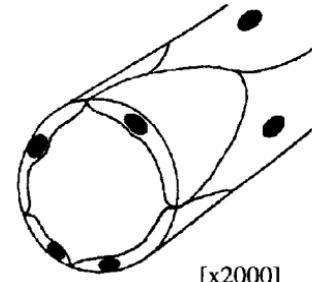
[x2]

Vein



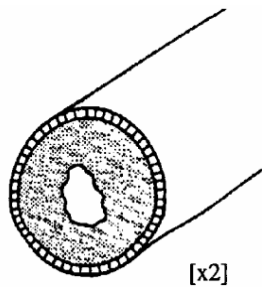
[x2]

Capillary

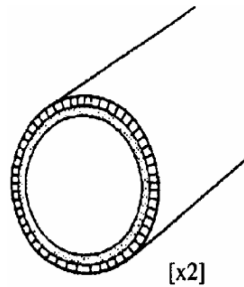


[x2000]

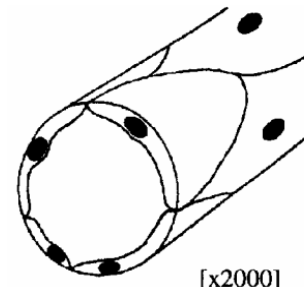
Artery



Vein



Capillary



Oxygenated or deoxygenated

AWAY or TOO heart

THICK or THIN walls

HIGH or LOW pressure

MUSCLE or NO MUSCLE

Name of artery: _____

VALVES or NO VALVES

LARGE or NARROW lumen

Oxygenated or deoxygenated

AWAY or TOO heart

THICK or THIN walls

HIGH or LOW pressure

MUSCLE or NO MUSCLE

Name of vein: _____

VALVES or NO VALVES

LARGE or NARROW lumen

Oxygenated or deoxygenated

AWAY or TOO heart

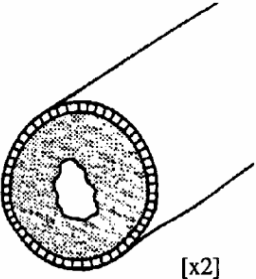
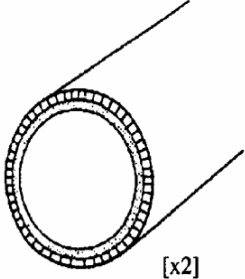
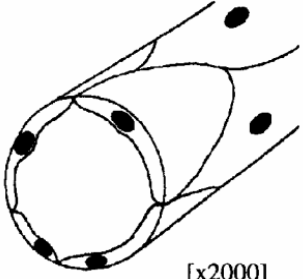
THICK or THIN walls

HIGH or LOW pressure

MUSCLE or NO MUSCLE

VALVES or NO VALVES

LARGE or NARROW lumen

Artery	Vein	Capillary
		
Away from heart or too heart Thick, thin, 1 cell thin walls High or low pressure Muscle or no muscle walls Example artery and vein Carry blood too or away from heart Large Lumen or small Valves or no valves		

LOb: Understand structures associated with the circulation of blood through the body.

Keywords: Arteries, capillaries, veins, valves, heart, circulation

Success Criteria:

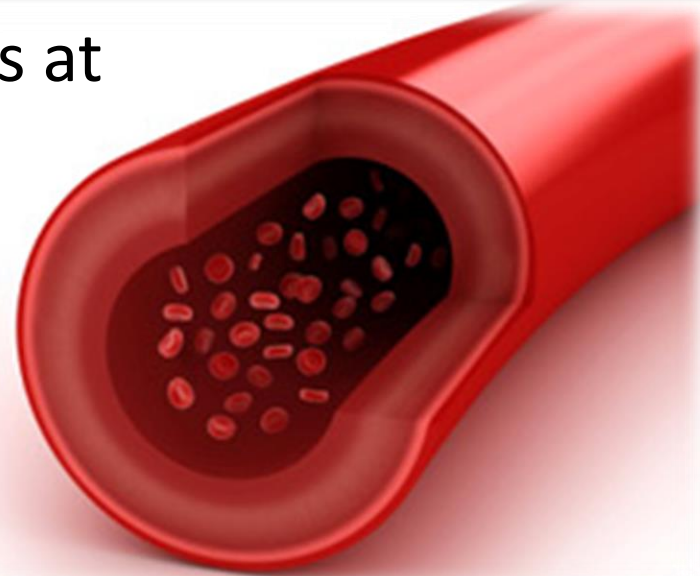
- ✓ Recall the structures of the blood vessels
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- Assess the importance of the double circulatory system



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Explaining Artery structure

The heart pumps blood through arteries at **high pressure.**



Thick muscular walls protects arteries from bursting.



Elastic walls allow arteries to stretch and expand with every heart beat.

Explaining Vein structure

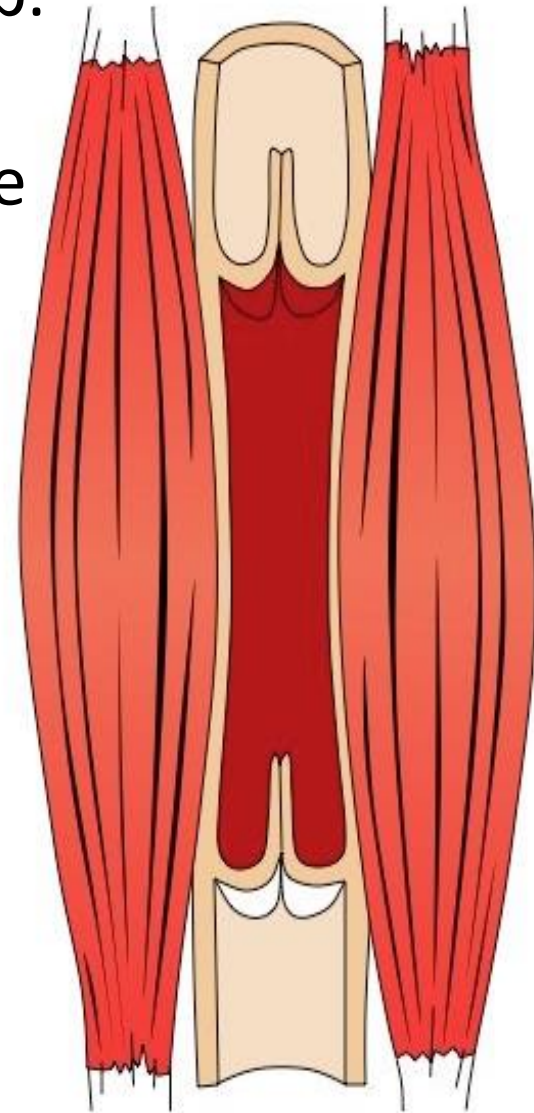
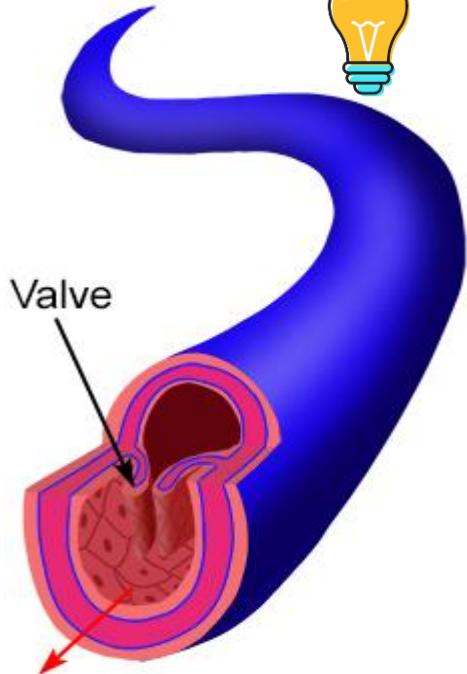
Blood returning to the heart is at such **low pressure** it can not make it back without help.



Muscles around veins squeeze veins pushing blood through.

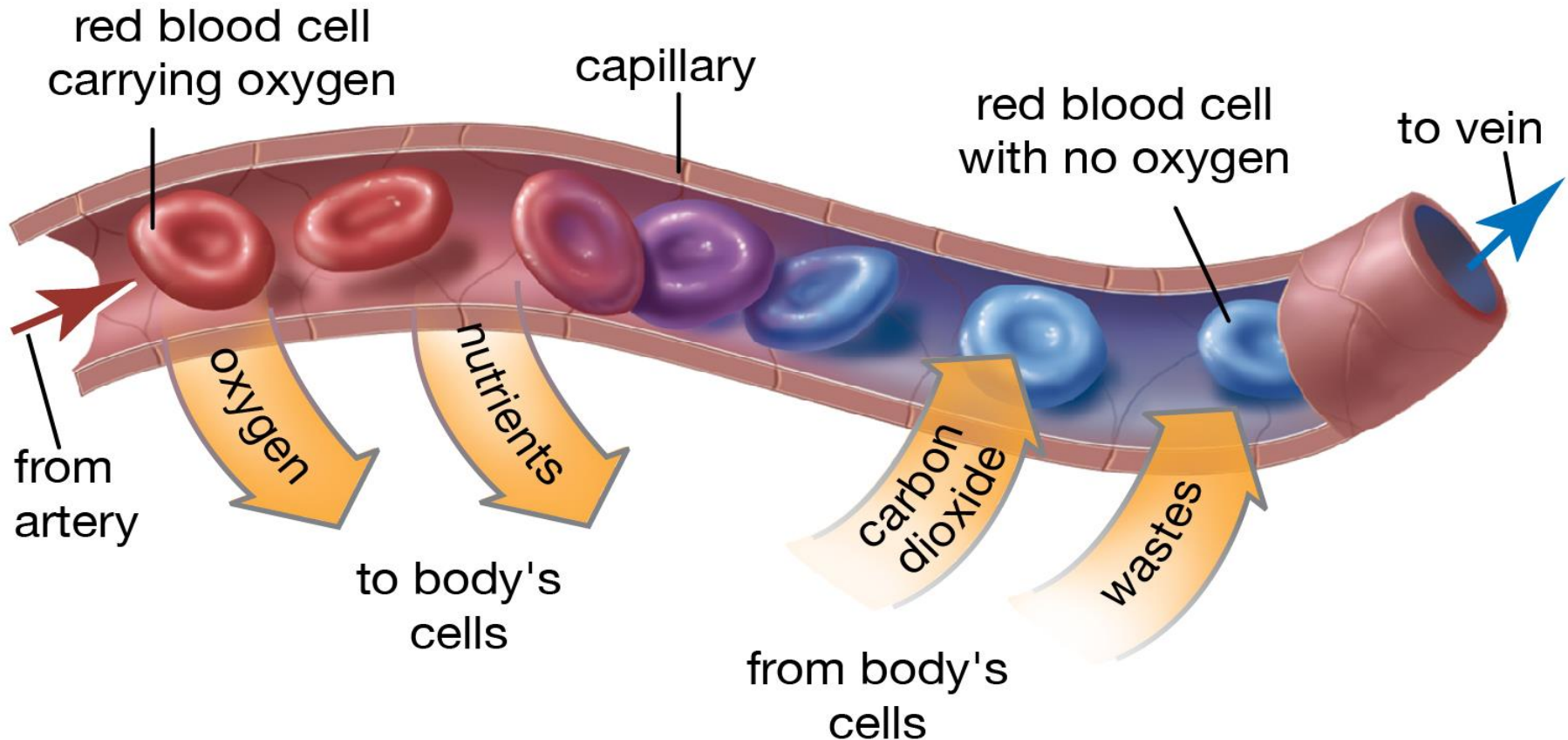


Valves stop the blood from going backwards.



Explaining Capillary Structure

Substances need to easily **diffuse** from the blood into the surround tissue and back again so the **walls need to be as thin as possible.**



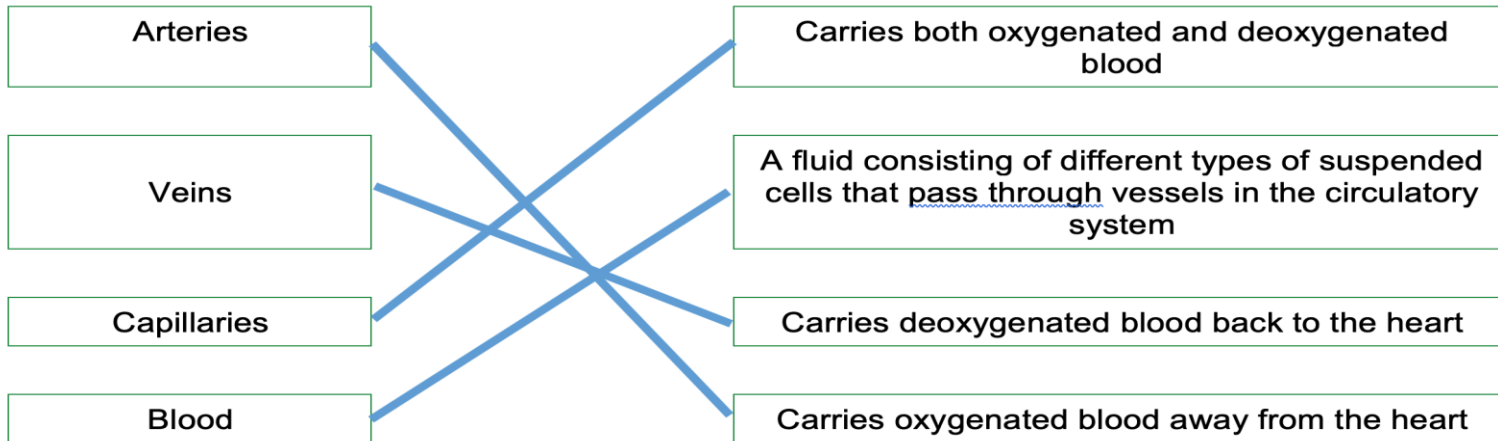
Activity: Complete the worksheet

**NOW IT'S
YOUR TURN**

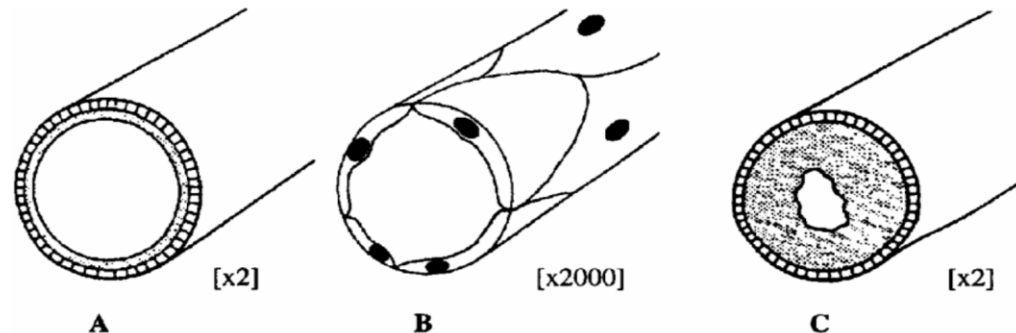
Review

Questions

1 a Link each vessel with its function.



b Name vessel C and describe two structures of this vessel.



Vessel C is an artery [1] It has thick layer of muscle [1] It is elastic and it can expand [1]

(3 mark)

b What is the function of vessel B. Give one adaptation for the function?

To allow substances to diffuse out of it into the surrounding tissues [1] Its walls are only one cell thick [1]

(2 mark)

Activity: Complete the worksheet

2 Answer these questions about veins

a Why do veins need valves?

Because blood is at a low pressure [1]

..... (1 mark)

b What do valves do?

Prevent blood from going backwards [1]

..... (1 mark)

c Describe why muscles are important for veins.

**Muscles press on the veins [1] This
pushes blood through them [1]**

.....
..... (2 mark)

LOb: Understand structures associated with the circulation of blood through the body.

Keywords: Arteries, capillaries, veins, valves, heart, circulation

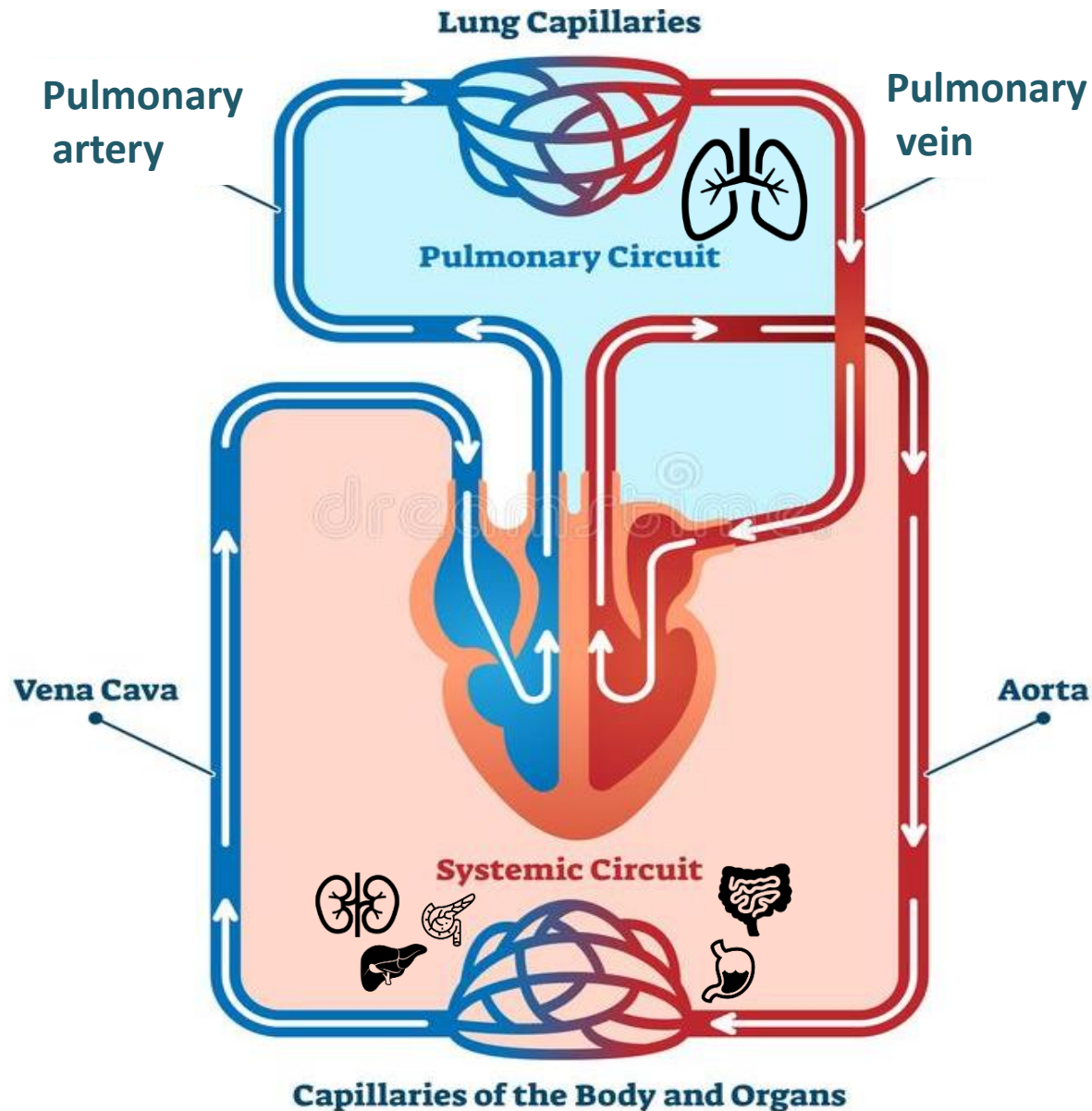
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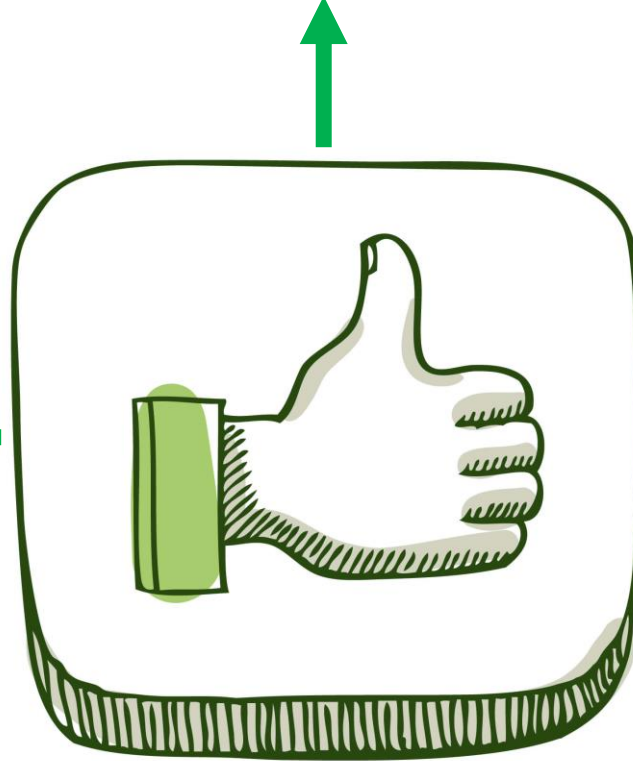


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Double circulatory system. Why is it called Double?



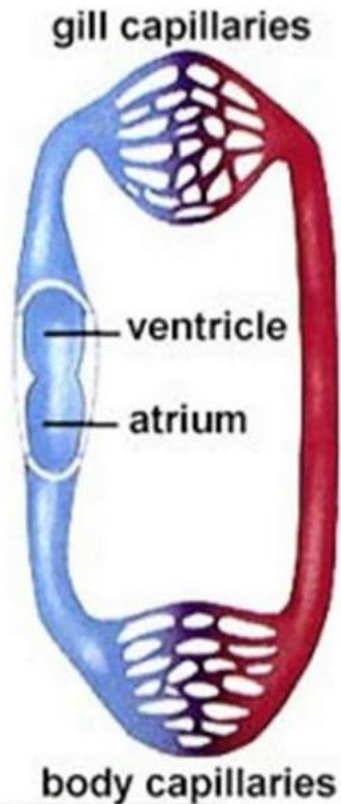
Allows **blood** to be **sent**
at **very high pressure**



Makes the
circulatory
system **very**
efficient

More areas of
the body
receive
oxygenated
blood

Fish have a single circulatory system. Blood goes from the heart to the gills, around the body then back to the heart. Describe the disadvantages of a single circulatory system for active land mammals. [4 marks]



Keywords

Large
Efficient
Move areas
Pressure

Review

Mammals are **larger**. Blood needs to travel further. A double circulatory systems allows blood to be pumped at **high pressure** so it can reach **more areas** of the body. The double circulatory system is also **more efficient** for the needs of land animals.

LOb: Understand structures associated with the circulation of blood through the body.

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keyfacts



Blood flows through vessels. **Arteries, capillaries** and **veins**.



Capillaries allow **substances to diffuse in and out** of the blood. They are **very narrow** and **thin walled = fast diffusion**



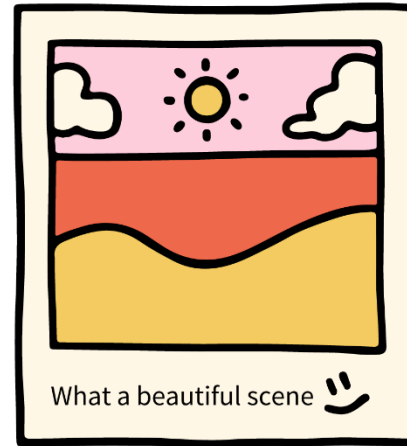
Valves in veins allow **blood** to **travel in one direction** under low pressure **preventing back flow**.



Human have a **double circulatory system**. This allow blood to travel at high pressure to more areas of the body.

Plenary: Describe your learning in

A picture
and
caption



Write a question and
it's answer

- — 3
- — Bullet
- — points